

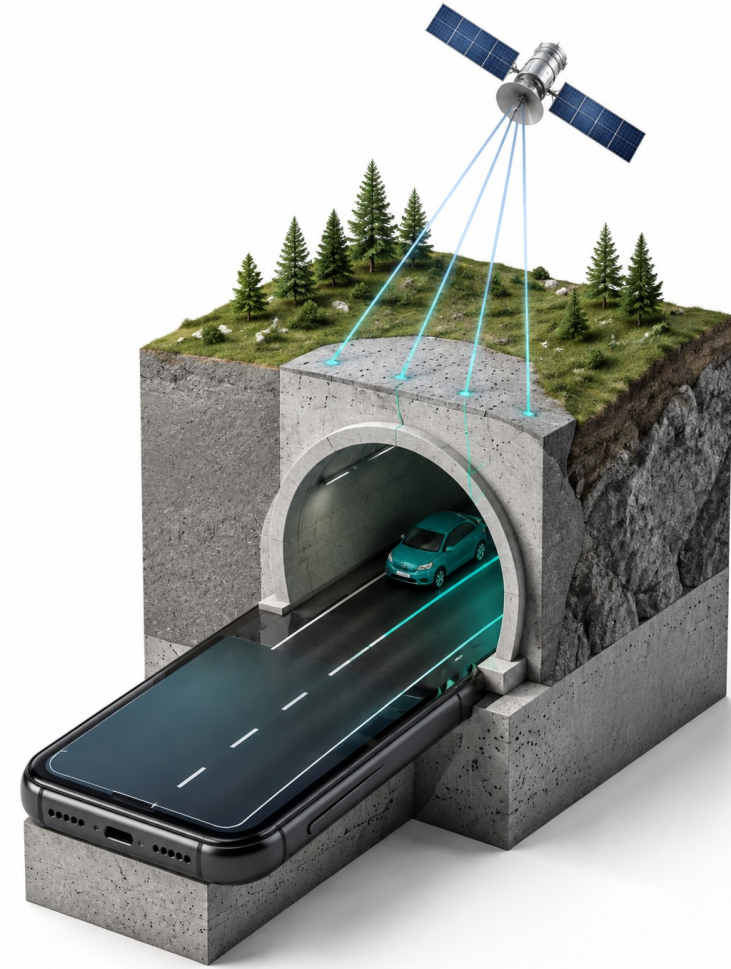
NavDR

Smartphone navigation through GNSS outages

AI-ML based Intelligent Dead Reckoning
system for seamless navigation

Problem Statement 26168

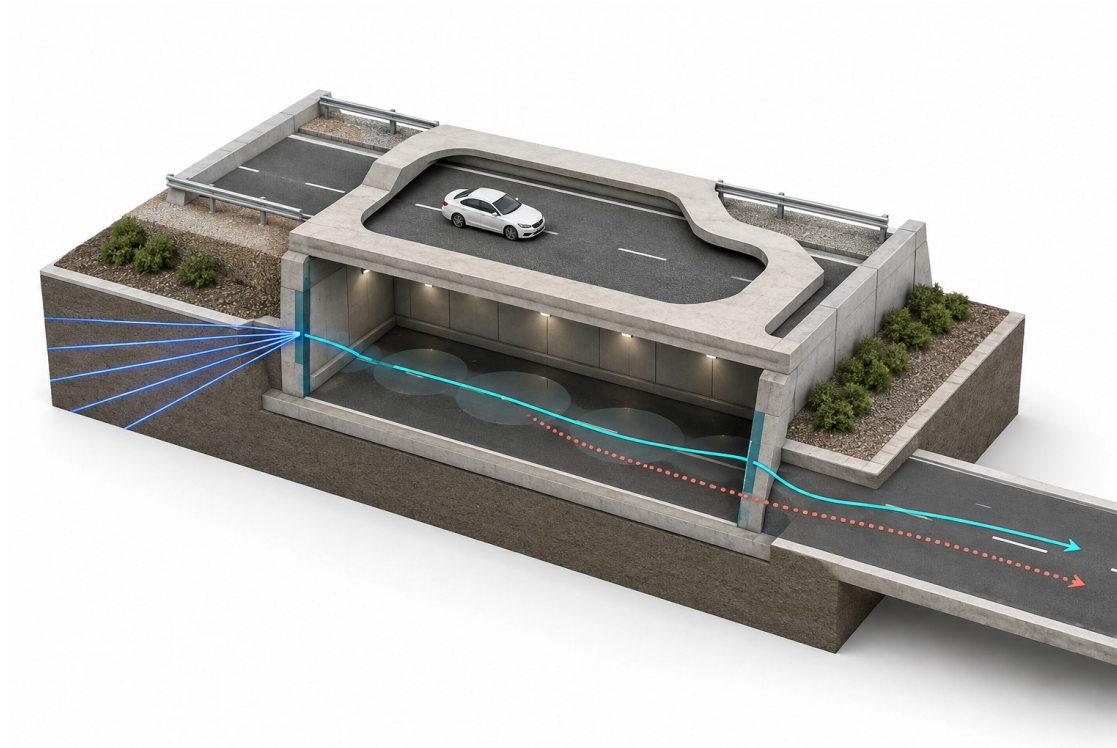
ISRO • Software • Miscellaneous



Concept illustration



Motion continuity when GNSS becomes unreliable



GNSS available

Dead reckoning

Recovery

THE PROBLEM

GNSS loses coverage.
IMU bias builds position error.

THE RESPONSE

Combine phone motion
sensors,
velocity estimates and road
constraints to bridge outages.

Validation target: peak outage error below 10% of outage distance



TECHNICAL APPROACH



Current: 100 Hz simulated IMU and replay

Next: trained weights and device validation

- 01 Phone IMU**
Acceleration and rotation
- 02 Frame alignment**
Fixed mounting configuration
- 03 TCN velocity**
49,665 parameters; untrained demo
- 04 Planar EKF**
State estimation and GNSS gating
- 05 Road HMM**
Road-constrained position output



Prototype to field validation



Upper stages planned

Risk	Response
Mounting changes	Detect and realign
GNSS multipath spike	Gate inconsistent fixes
Late IMU samples	Flag timestamp gaps
Unfamiliar routes / phones	Evaluate held-out data

Working now

Console, sensor simulation, EKF, road matching and replay



IMPACT AND BENEFITS

Navigation continuity using an existing phone

Drivers

Position awareness through short GNSS outages

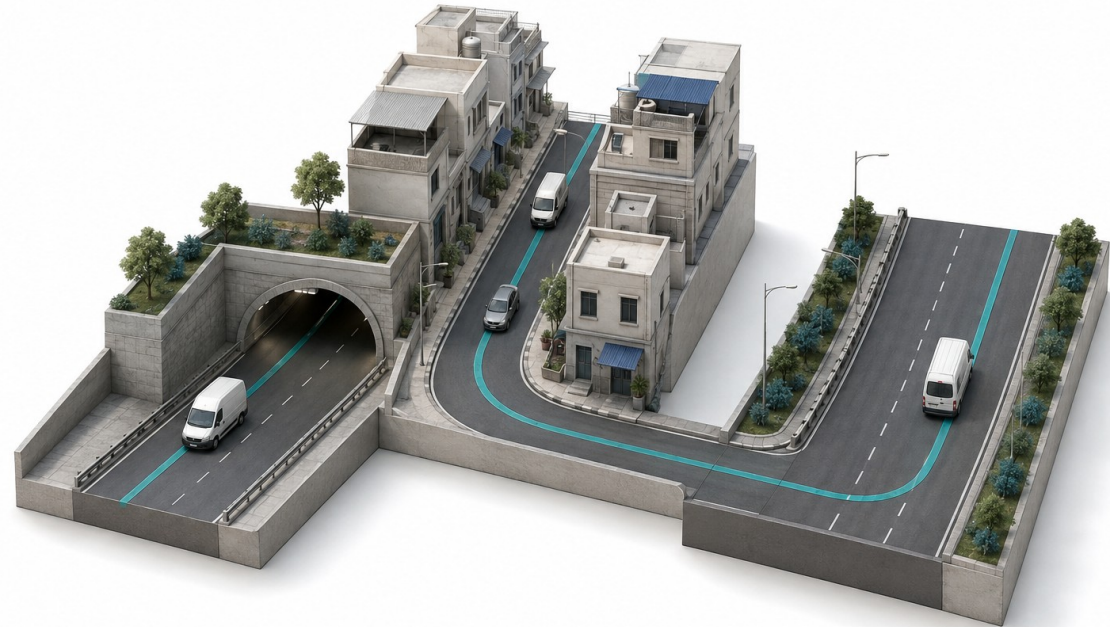
Fleets

Route continuity through tunnels and dense streets

Deployment goal

No dedicated vehicle sensor hardware

Measure error, recovery time, latency and phone energy use





RESEARCH AND REFERENCES

[IO-VNBD](#)

Vehicle and smartphone motion data

[AI-IMU Dead-Reckoning](#)

Learning-assisted inertial estimation

[Temporal Convolutional Networks](#)

Causal sequence modelling

[Valhalla Meili](#)

HMM map matching

[Android GnssStatus](#)

Satellite signal observations

Synthetic prototype. TCN untrained. Field validation pending.

[YouTube Demo](#)

youtu.be/GoU52CqyaNI

[GitHub Repository](#)

github.com/Sudhanshu-01-ui/gps

